Econometrics 2 capstone proposal code

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Importing and cleaning data

Tables used for this proposal:

- Psychology (S10AI)
- Housing: water (S12AI)
- Housing: sewage facilities (S12AII)

Importing the Pyschology table

```
##############################################################################
################################ PSYCHOLOGY TABLE ##############################
##############################################################################

s10ai <- read_csv("data/S10AI.csv") %>%
  select(hhno, hhmid, depression, sex = s1d_1, age = s1d_4i) %>%

  #Creating a unique household id column
  mutate(id = hhno + hhmid) %>%

  #Creating a new column as our depression_dummy. Kessler scores between 10-19 have a score o
f one in the data (== "likely to be well"). Anyone with scored higher than this has a score >
1, which classifies them as likely to have at least a mild disorder.
  mutate(depression_dummy = case_when(

    depression > 1 ~ 1, # Depressed
    TRUE ~ 0 # Not depressed

  ))
```

Importing the housing tables

In the following order:

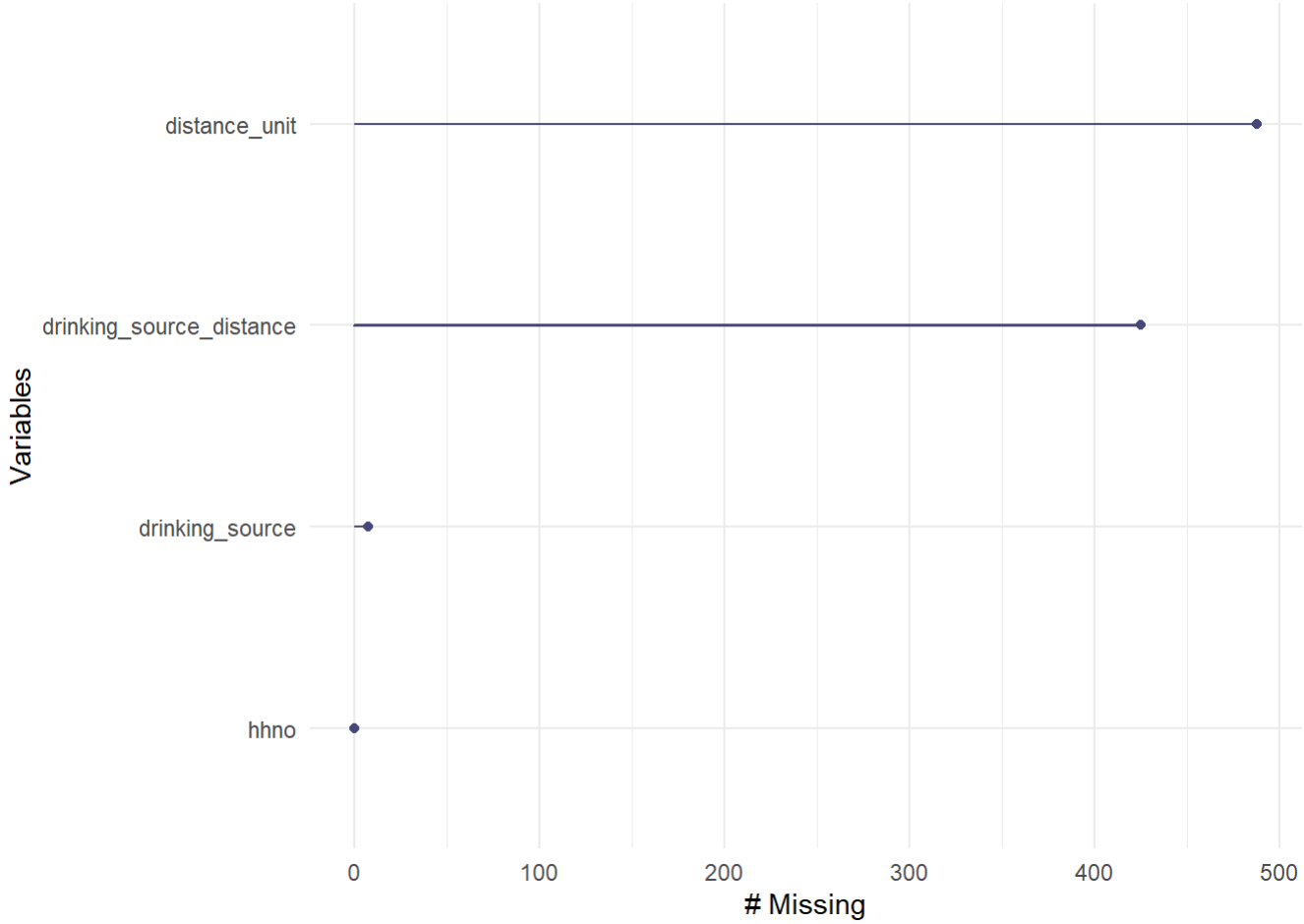
- Water
- Sewerage

```
################################################################################
############################### HOUSING TABLES ################################
################################################################################

############################### WATER TABLE ####################################

s12ai <- read_csv("data/S12AI.csv") %>%
  select(hhno,drinking_source = s12a_9i, drinking_source_distance = s12a_10ai, distance_unit
= s12a_10aii) %>%

  #Editing the drinking_source_distance cells to make them all the same scale: metres.

  mutate(drinking_source_distance = case_when(

    distance_unit == 0 ~ 0,  # In house
    distance_unit == 1 ~ 0,  # In yard (assuming 0 meters)
    distance_unit == 2 ~ as.numeric(drinking_source_distance),  # Already in meters
    distance_unit == 3 ~ as.numeric(drinking_source_distance) * 1000,  # Kilometers to meters
    distance_unit == 4 ~ as.numeric(drinking_source_distance) * 1609.344,  # Miles to meters
    TRUE ~ drinking_source_distance

  ))
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## Rows: 4972 Columns: 72
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr  (2): s12a_15, s12a_15i
## dbl (67): id1, id3, id4, id2, s12a_1, s12a_2i, s12a_2ii, s12a_2iii, s12a_3, ...
## lgl  (3): s12a_4i, s12a_4ii, s12a_4iii
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
gg_miss_var(s12ai)
```



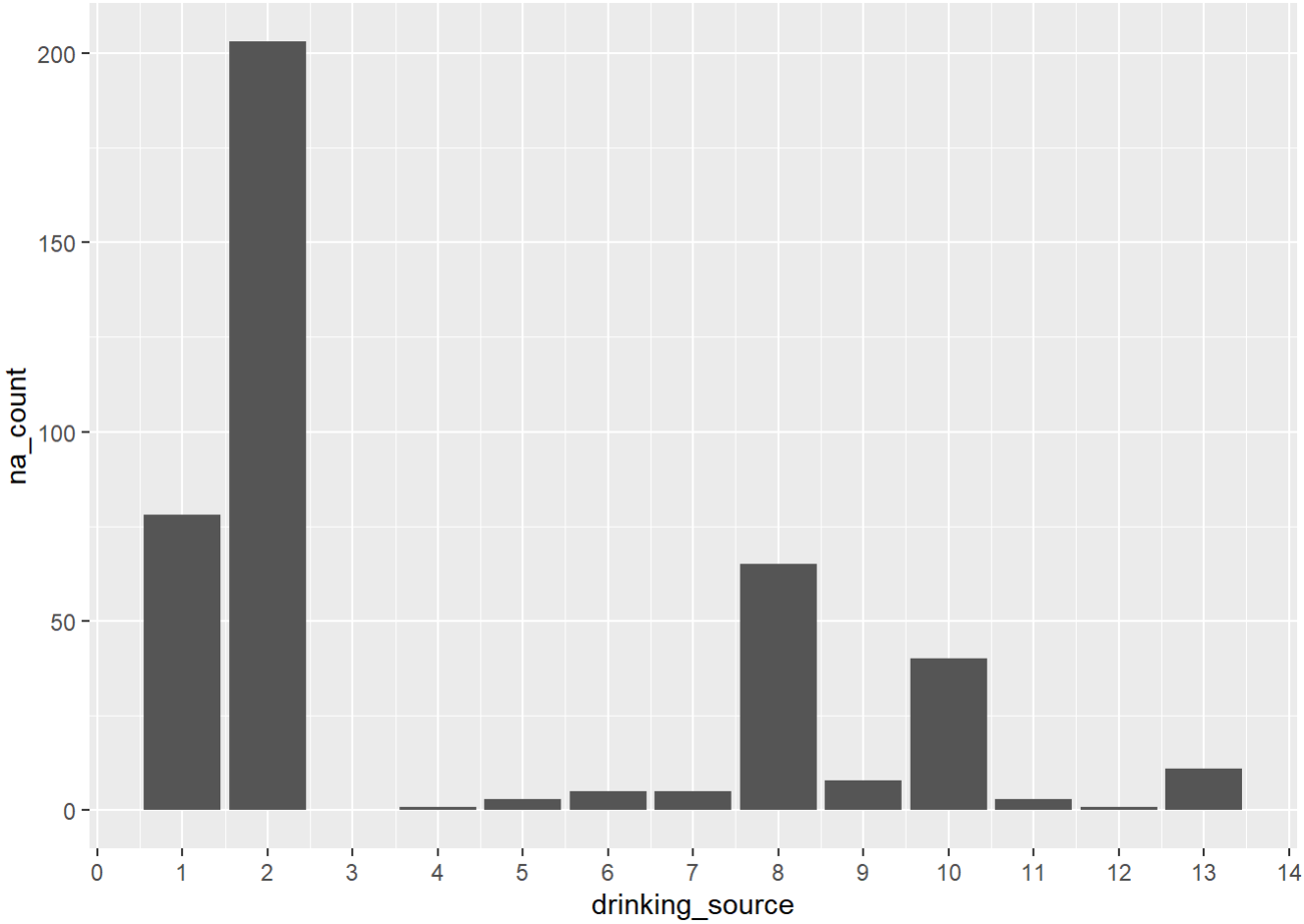
The chart above shows us that there is a lot of missing values for distance_unit. This likely have something to do with the drinking source of each household. I need to collect all the NA data together in order to diagnose the problem.

The chart below shows us that:

- Most of the problem is in 1 and 2, which corresponds to plumbing in the house. We can change their distances to zero.
- 8 is also a clear problem, which is bottled water. Therefore leave this as NA.
- 9 and 10 are protected wells and boreholes. Without more information about how far away they are (unavailable) we need to leave these as NAs.

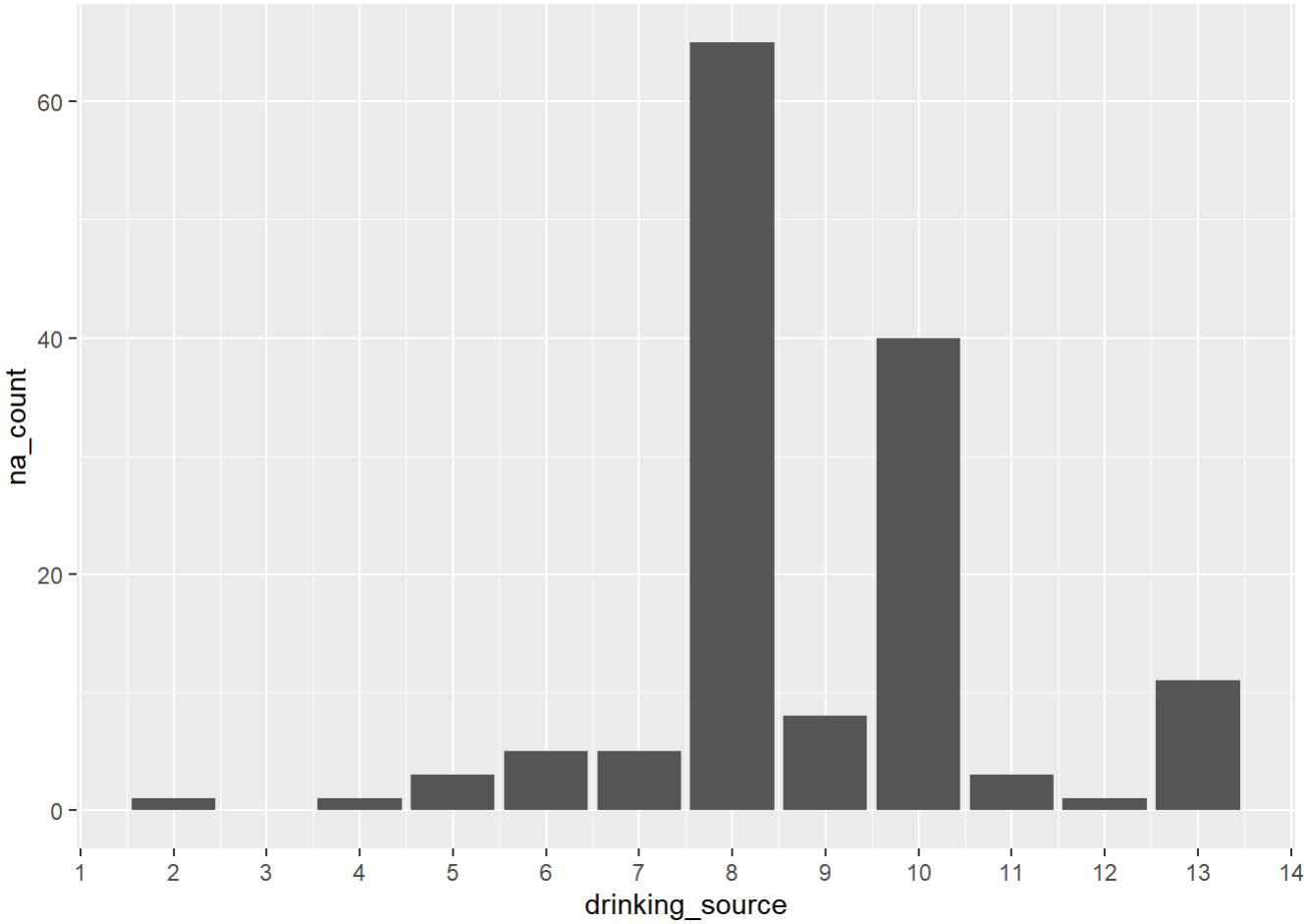
```
# Extracting and charting NA data

na_data <- s12ai %>%
  filter(is.na(drinking_source_distance)) %>%
  group_by(drinking_source) %>%
  summarise(na_count = n())

ggplot(na_data, aes(x = drinking_source, y = na_count)) +
  geom_bar(stat = "identity") +
  scale_x_continuous(breaks = scales::pretty_breaks(n = 14))
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_bar()`).
```




```
# Now I have diagnosed the problem, I need to make the necessary changes to the dataframe suc
h that dirnking_sources with values 1 and 2 have a distance of zero. All other NAs remain giv
en data limitations.

s12ai <- s12ai %>%
  mutate(drinking_source_distance = case_when(

    is.na(distance_unit) & drinking_source %in% c(1, 2) ~ 0,
    TRUE ~ drinking_source_distance
  ))




# Repeating the NA value analysis/chart below, the scale are now sufficiently small to contin
ue/we don't have any other information that could help reduce the incidence of NAs.

na_data <- s12ai %>%
  filter(is.na(drinking_source_distance)) %>%
  group_by(drinking_source) %>%
  summarise(na_count = n())

ggplot(na_data, aes(x = drinking_source, y = na_count)) +
  geom_bar(stat = "identity") +
  scale_x_continuous(breaks = scales::pretty_breaks(n = 14))
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_bar()`).
```

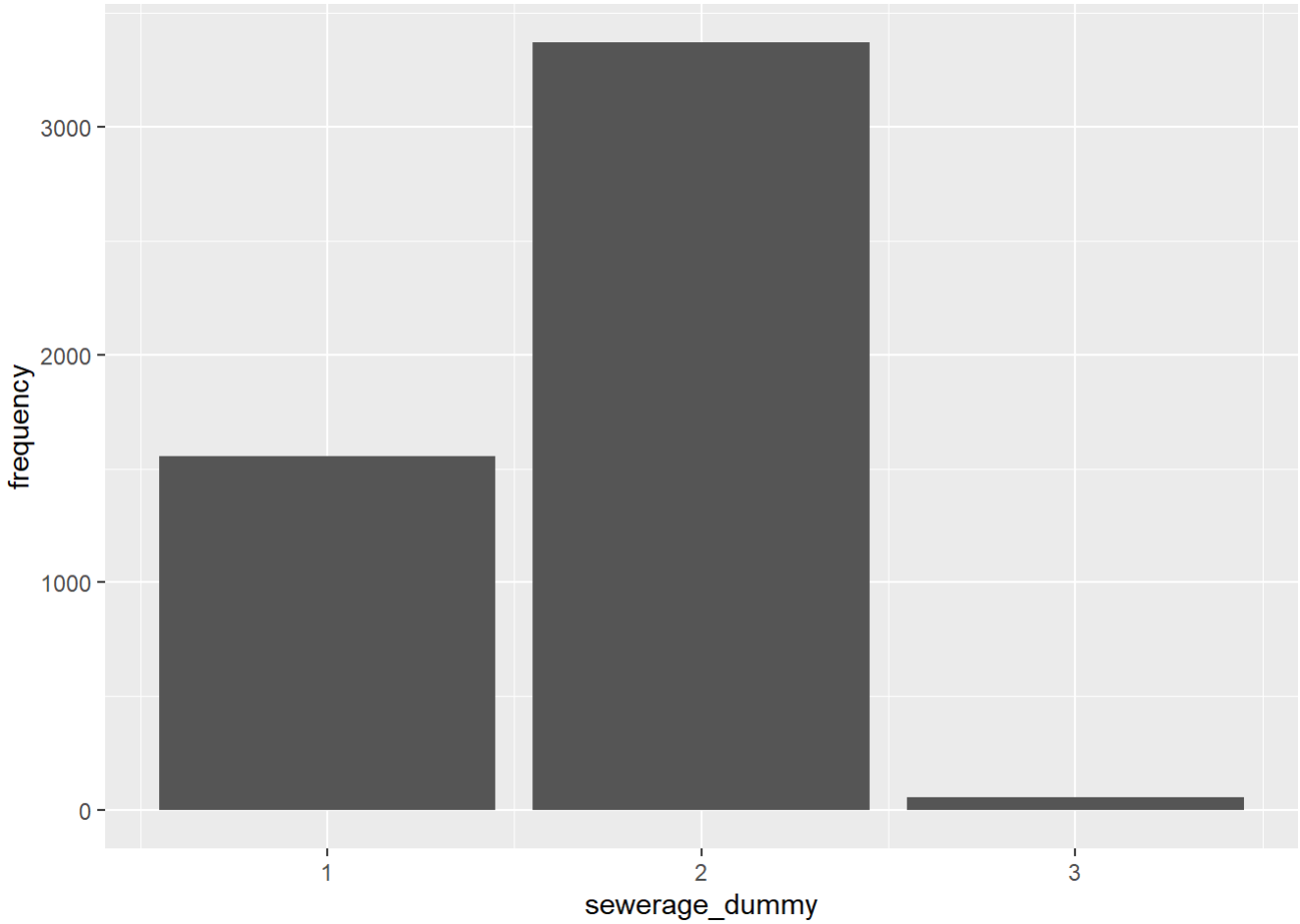



```
################################ SEWERAGE TABLE ##################################

s12aii <- read_csv("data/S12AII.csv") %>%
  select(hhno, sewerage_dummy = s12b_23)
```

```
## Rows: 4998 Columns: 30
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## dbl (30): id1, id2, id3, id4, s12b_1, s12b_2, s12b_3, s12b_4, s12b_5, s12b_6...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
  # Is there any open sewer,drain in/around the house? Note: 1 == Yes, 2 == No, 3 == Drains a
re covered

frequency_test <- s12aii %>%
    group_by(sewerage_dummy) %>%
  summarise(frequency = n())

ggplot(frequency_test, aes(x = sewerage_dummy, y = frequency)) + geom_bar(stat = "identity")
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_bar()`).
```



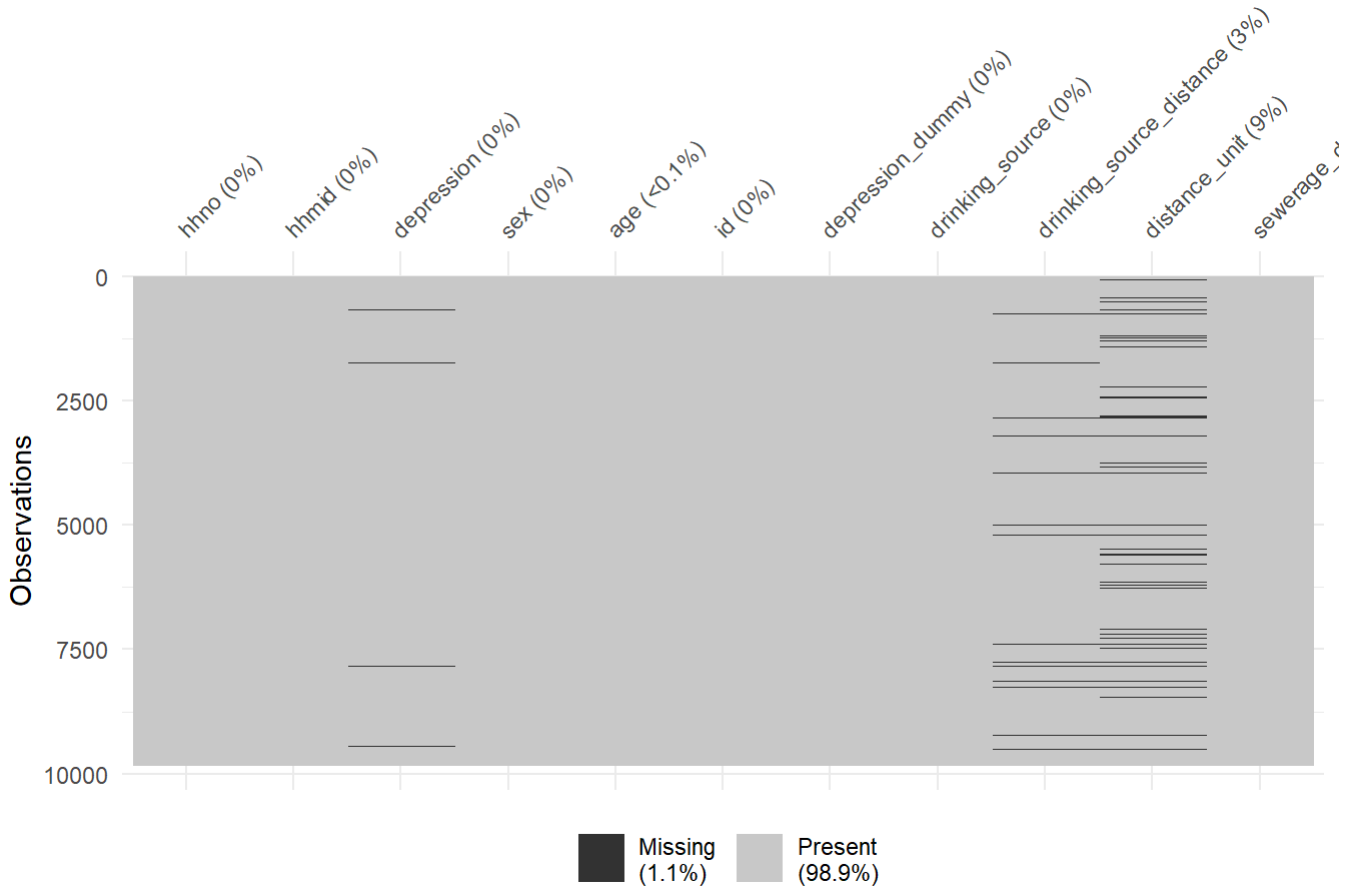

```
# Given how infrequently option 3 occurs, we are going to exclude it to ensure our variable a
ctually as a dummy.

s12aii <- s12aii %>%
  filter(sewerage_dummy != 3)
```

Joining data

Household data is not provided at the individual level. Therefore, we need to append it to our psychological data.

Doing a quick NA visualisation I can see that there are a few columns with NA values. Given how small they are as proportions, I omit the NA values for depression and drinking_source_distance. I don't both with distance_unit (its only use was to help us clean the data earlier.)

```
data <- s10ai %>%
  inner_join(s12ai, by = "hhno") %>%
  inner_join(s12aii, by = "hhno")

vis_miss(data)
```




```
# Omitting the relevant NAs

data <- data %>%
  filter(!is.na(depression)) %>%
  filter(!is.na(drinking_source_distance))
```

Creating summary statistics

```
summary_stats <- data %>%
  summarise(
    mean_depression_dummy = mean(depression_dummy, na.rm = TRUE),
    sd_depression_dummy = sd(depression_dummy, na.rm = TRUE),
    min_depression_dummy = min(depression_dummy, na.rm = TRUE),
    max_depression_dummy = max(depression_dummy, na.rm = TRUE),

    mean_drinking_source_distance = mean(drinking_source_distance, na.rm = TRUE),
    sd_drinking_source_distance = sd(drinking_source_distance, na.rm = TRUE),
    min_drinking_source_distance = min(drinking_source_distance, na.rm = TRUE),
    max_drinking_source_distance = max(drinking_source_distance, na.rm = TRUE),

    mean_age = mean(age, na.rm = TRUE),
    sd_age = sd(age, na.rm = TRUE),
    min_age = min(age, na.rm = TRUE),
    max_age = max(age, na.rm = TRUE),

    mean_sex = mean(sex, na.rm = TRUE),
    sd_sex = sd(sex, na.rm = TRUE),
    min_sex = min(sex, na.rm = TRUE),
    max_sex = max(sex, na.rm = TRUE),

    mean_sewerage_dummy = mean(sewerage_dummy, na.rm = TRUE),
    sd_sewerage_dummy = sd(sewerage_dummy, na.rm = TRUE),
    min_sewerage_dummy = min(sewerage_dummy, na.rm = TRUE),
    max_sewerage_dummy = max(sewerage_dummy, na.rm = TRUE)
  )

# Reshape the summary statistics data frame
summary_stats <- summary_stats %>%
  pivot_longer(
    everything(),                      # Select all columns to reshape
    names_to = c("statistic", "variable"),  # Create two new columns: statistic and variable
    names_sep = "_"                    # Split the names at the underscore
  ) %>%
  select(variable, statistic, value) %>%
  mutate(value = round(value,2)) %>%
    pivot_wider(
    names_from = statistic,             # Use statistic names as new column names
    values_from = value                  # Fill with the values from the value column
  )
```

```
## Warning: Expected 2 pieces. Additional pieces discarded in 12 rows [1, 2, 3, 4, 5, 6, 7,
## 8, 17, 18, 19, 20].
```

```
print(summary_stats)
```

```
## # A tibble: 5 × 5
##   variable      mean       sd   min    max
##   <chr>        <dbl>    <dbl> <dbl>  <dbl>
## 1 depression    0.31     0.46     0      1
## 2 drinking   9643.   59909.       0 800000
## 3 age          39.0     18.7      1    109
## 4 sex           1.55     0.5      1      2
## 5 sewerage      1.67     0.47     1      2
```

```
write_csv(summary_stats, "summary_stats.csv")
```